Propelling Through Water Resistance

Have you ever tried to run in a swimming pool, ocean, or lake? You can't move nearly as fast as you can on the beach or grass. It seems like you're moving in slow motion. You've encountered water resistance.

Water resistance is a force that competes against motion. It is similar to air resistance, but it is much stronger. Air is much lighter and thinner than water. When you push through air, it moves aside much more quickly than water does. Have you ever dropped water goggles in deep water? They fall much more slowly than they would fall the same distance through air. The force of **buoyancy** pushes up on an object in the water, but if the object is heavy for its size, gravity wins and the object slowly sinks.

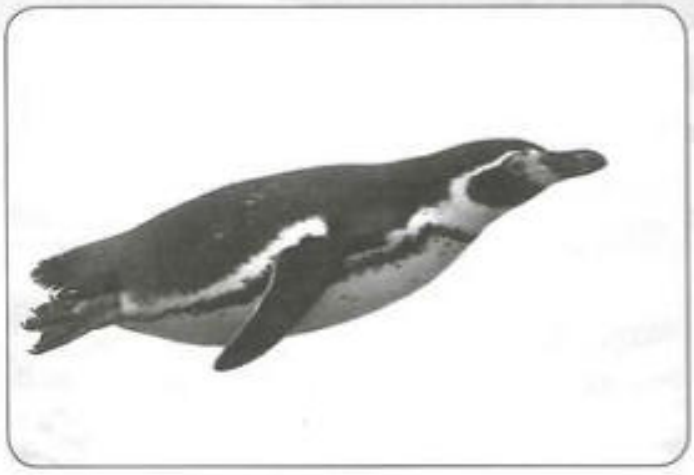
While air and water treat moving objects differently, there are some similarities. When objects fall, they must push the water or air aside. The larger the area that is pushing down, the more slowly it falls. For example, if you drop a comic book with the cover facing down, it will drop a little more slowly than if the spine, or edge, of the comic book is facing down. This is true for both water and air.

So how can we **propel**, or move, something through water? Isaac Newton, a scientist from the 1600s, helped us with that. Newton described how forces move and stop objects and change the way objects move. He noticed that when you push or pull something and make it move, it will react by moving with the same amount of force the opposite way. For example, if you push a basketball toward the ground, the ball will move away from your hand. When it hits the ground, it bounces back up in the air. Swimmers push water from in front of them to behind them, which propels the water back and the swimmer forward.

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STEM Challenge:
Paddle Boat

Visual Literacy



Some birds swim to catch fish, like this penguin. It shoots forward under the water every time it pushes its wings backward.



When people swim, they pull themselves through the water with their arms and push themselves from behind by kicking their feet.



Boats often use paddles to move forward. The rower operates the paddles on a rowboat, canoe, or kayak. On a paddlewheel boat, the paddles are attached to a turning wheel, either on the back or sides of the boat.

Paddle Boat

Challenge: Design a paddle boat that moves forward in water.

Testable goal: The boat will move across the water at least 1 foot (30 centimeters).

Research: Look at pictures of paddle boats. Notice how the boats are constructed. Think about how science, technology, engineering, and math are used to create a paddle boat.



Brainstorm: Draw one or more design ideas for a paddle boat. There are many different ways to complete this challenge. Be creative!

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STEM Challenge:

Paddle Boat

Suggested Materials List

Item for each group

- [ ] scissors

Items for the whole class

- [ ] empty milk or juice carton
- [ ] empty rectangular margarine containers
- [ ] large rubber bands
- [ ] string
- [ ] tongue depressors
- [ ] thin dowels
- [ ] plastic or metal spoons
- [ ] rubber cement
- [ ] hole puncher
- [ ] _____
- [ ] _____
- [ ] _____
- [ ] _____
- [ ] _____
- [ ] _____
- [ ] _____
- [ ] _____
- [ ] _____

Item for testing

- [ ] large tub of water

Paddle Boat

1 Plan

Think about the materials available to you. How can you use the materials to create a paddle boat?

2 Create

Use the materials you have to create a paddle boat. Follow your design as closely as possible.

3 Test

Place your paddle boat in water and make it go. Does it move forward in water? Does it stay upright?

4 Evaluate and Revise

Evaluate the performance of your paddle boat. What revisions can you make to improve its performance?

Repeat the design process until you are satisfied with your paddle boat.



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STEM Challenge:

Paddle Boat

Design Process

1 Plan: Describe and draw your design. List your materials.



2 Create



3 Test: Describe and draw your results.



4 Evaluate

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STEM Challenge:

Paddle Boat

Redesign Process

1 Redesign: Describe what you will change. Draw your new design.



2 Revise or re-create



3 Test: Describe and draw your results.



4 Evaluate

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